Microeconomics I

Class 7 — The Slutsky equation and the money value of a price change

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20 October 2026

Short quiz

8 minutes · pens down at 10:08

Circle the letter of your answer in the block at the top of the sheet.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

Plan for today

10:00	Short quiz
10:10	Warm-up; a mug, three prices
10:20	The Slutsky equation: substitution and wealth effects
10:35	Giffen goods, revisited
10:40	In pairs: gross and net cross effects
10:50	Break
11:00	The Slutsky matrix; the map of x, v, h, e
11:10	Compensating and equivalent variation (CV, EV)
11:29	In pairs: a cheaper good
11:38	Deadweight loss; summary

Practicalities

- Problem set 2: due at the start of class, Wed 21 Oct — on paper, one copy per group
- Office hours: Wednesdays 14:00–15:00, Room 445

  up to 20 minutes per student; those who book by e-mail have priority
- Slides and reading: pkocourek.com/teaching/micro-1 · pkocourek@ub.edu

Warm-up: always true, or not?

Last class. True for every consumer? One minute with your neighbour.

p_1 rises $\Rightarrow$
$h_1(p, u)$ does not rise

p_1 rises $\Rightarrow$
$x_1(p, w)$ does not rise

$e(p, u)$ is homogeneous of degree 0 in p

$e\Big(p, v(p, w)\Big) = w$

Class 6, slides 10, 13, 14; class 5, slide 22. MWG Prop. 3.E.2, 3.E.4; (3.E.1).

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true: compensated law of demand

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false: Giffen (class 5's rice)

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$$e\Big(p, v(p, w)\Big) = w$$

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$h_1(p, u)$ does not rise

true: compensated law of demand

p_1 rises $\Rightarrow$
$x_1(p, w)$ does not rise

false: Giffen (class 5's rice)

$e(p, u)$ is homogeneous of degree 0 in p

false: degree 1

$$e\Big(p, v(p, w)\Big) = w$$

Class 6, slides 10, 13, 14; class 5, slide 22. MWG Prop. 3.E.2, 3.E.4; (3.E.1).

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p_1 rises $\Rightarrow$
$h_1(p, u)$ does not rise

true: compensated law of demand

p_1 rises $\Rightarrow$
$x_1(p, w)$ does not rise

false: Giffen (class 5's rice)

$e(p, u)$ is homogeneous of degree 0 in p

false: degree 1

$$e\Big(p, v(p, w)\Big) = w$$

true: duality

h_1 never rises with p_1; x_1 can. Today: the difference is a wealth effect.

Class 6, slides 10, 13, 14; class 5, slide 22. MWG Prop. 3.E.2, 3.E.4; (3.E.1).

A mug, three groups: whose prices should be equal?

Simon Fraser University: 77 students, a coffee mug, three groups at random

Sellers are given a mug: the lowest price at which they sell it

Buyers: the highest price they pay for one

Choosers: a mug or cash: the cash that is worth as much as the mug

Which two medians should standard theory make equal?

buyers and choosers · choosers and sellers · buyers and sellers

Kahneman, Knetsch & Thaler (1990), Journal of Political Economy 98(6), experiment 6.

Choosers and sellers face the same choice: 3.12 vs 7.12

Buyers	Choosers	Sellers
$2.87	**$3.12**	**$7.12**
pay at most	cash worth a mug	accept at least

medians, 77 students

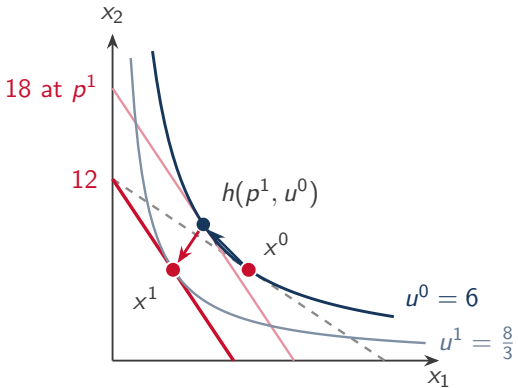
Choosers and sellers both choose between a mug and cash, yet sellers ask more than twice as much. What theory allows: after the break.

Kahneman, Knetsch & Thaler (1990), "Experimental tests of the endowment effect and the Coase theorem", JPE 98(6): 1325–1348, experiment 6, p. 1339.

The Slutsky equation

MWG 3.G

Good 1 dearer: x_1 falls $3 \to 2$ along u^0, then $2 \to \frac{4}{3}$ from lost wealth



x_1: $3 \to 2 \to \frac{4}{3}$

substitution -1, wealth $-\frac{2}{3}$

x_2: $2 \to 3 \to 2$

substitution $+1$, wealth -1

18 euros at p^1 keep her on u^0

Substitution: along u^0, at the new prices.

Wealth: the parallel shift from 18 euros down to her 12.

MWG 3.G, pp. 71–72; class 6, slide 14. $u = x_1 x_2$, $w = 12$, $p^0 = (2, 3) \to p^1 = (4.5, 3)$.

Slutsky equation: compensated slope = market slope + wealth term

$$\frac{\partial h_\ell(p,u)}{\partial p_k} = \frac{\partial x_\ell(p,w)}{\partial p_k} + \frac{\partial x_\ell(p,w)}{\partial w}\, x_k(p,w) \quad \text{at } u = v(p,w)$$

why: differentiate $h_\ell(p,u) = x_\ell\big(p, e(p,u)\big)$ in p_k; $\partial e/\partial p_k = h_k = x_k$ (Shephard)

$$\frac{\partial x_\ell}{\partial p_\ell} = \underbrace{\frac{\partial h_\ell}{\partial p_\ell}}_{\leq 0\text{: substitution}} \underbrace{- x_\ell \frac{\partial x_\ell}{\partial w}}_{\text{wealth}}$$

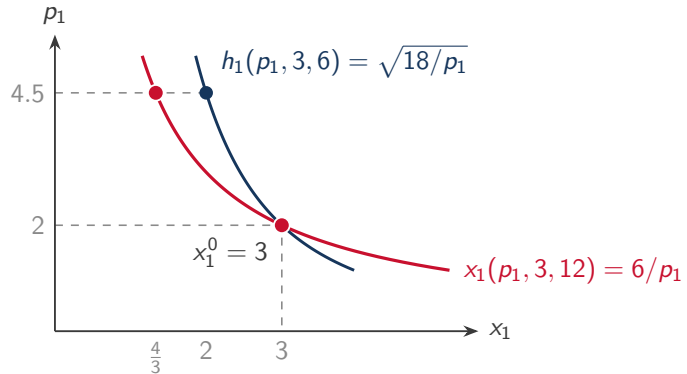
$x_1 x_2$ at $p = (2,3)$, $w = 12$: $-\frac{3}{2} = -\frac{3}{4} - 3 \cdot \frac{1}{4}$

A normal good falls more in market demand than in compensated demand.

MWG Prop. 3.G.3, p. 71, (3.G.3)–(3.G.4): u continuous, $\succsim$ locally non-satiated and strictly convex.

Market demand falls faster than compensated: good 1 is normal

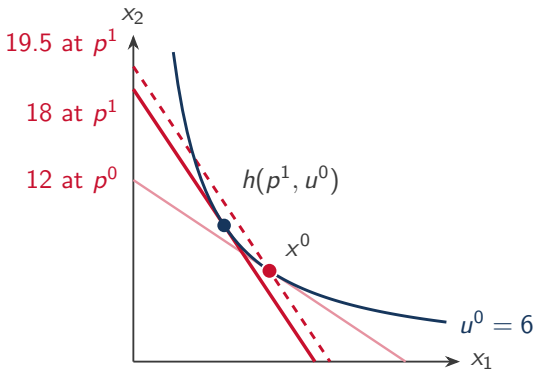
$u = x_1 x_2$, $p_2 = 3$, $w = 12$, $u^0 = 6$: both curves pass through $x_1 = 3$ at $p_1 = 2$



At $p_1 = 4.5$: compensated demand 2, market demand $\frac{4}{3}$; the gap is the wealth effect.

MWG Figure 3.G.1(a), p. 71; text p. 72.

Two ways to compensate: keep u^0 (6 euros) or x^0 affordable (7.5)



Hicks: $e(p^1, u^0) - w = 18 - 12 = 6$

back on her old curve

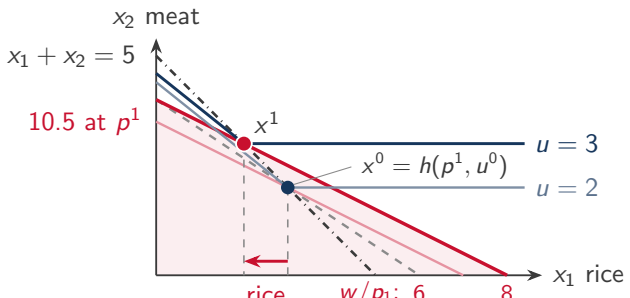
Slutsky: $p^1 \cdot x^0 - w = 19.5 - 12 = 7.5$

her old bundle affordable again

Hicks $\leq$ Slutsky, equal to first order: so the compensated slopes of slide 9 carry Slutsky's name.

MWG 3.G, pp. 72–73, Figure 3.G.2; class 6, slide 22.

Giffen needs inferior: class 5's rice, $3 \to 3 \to 2$



rice p_1: $2 \to 1.5$

x_1: $3 \to 3 \to 2$

substitution 0 (the kink on the food line $x_1 + x_2 = 5$), wealth -1

at $p_1 = 2$:

$$\frac{\partial x_1}{\partial p_1} = 0 - 3 \cdot (-1) = +3$$

$\partial h_1/\partial p_1 = 0$, $x_1 = 3$, $\partial x_1/\partial w = -1$

Substitution is never positive, so $\partial x_\ell/\partial p_\ell > 0$ needs $\partial x_\ell/\partial w < 0$: Giffen needs inferior.

Class 5, slides 19 and 22: $u = \min\{x_2,\ x_2 - 4(5 - x_1 - x_2)\}$, $w = 12$, $p_2 = 3$. MWG Prop. 3.G.3, p. 71.

In pairs: net cross effect = gross + wealth term

Class 6, slide 21, at its start: $p = (2, 3)$, $w = 12$. x_2 from the budget. $\sqrt{x_1} + \sqrt{x_2}$: the CES with $\rho = \frac{1}{2}$.

u	$x_1(p, w)$	gross $\frac{\partial x_1}{\partial p_2}$	wealth term $x_2 \frac{\partial x_1}{\partial w}$	net $\frac{\partial h_1}{\partial p_2}$
$\min\{x_1, x_2\}$	$\frac{w}{p_1 + p_2}$			
$x_1 x_2$	$\frac{w}{2p_1}$			
$\sqrt{x_1} + \sqrt{x_2}$	$\frac{w\, p_2}{p_1(p_1 + p_2)}$			

Four minutes.

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$\min\{x_1, x_2\}$	$\frac{w}{p_1 + p_2}$	-0.48	$+0.48$	0
$x_1 x_2$	$\frac{w}{2p_1}$			
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$\min\{x_1, x_2\}$	$\frac{w}{p_1 + p_2}$	-0.48	$+0.48$	0
$x_1 x_2$	$\frac{w}{2p_1}$	0	$+0.5$	0.5
$\sqrt{x_1} + \sqrt{x_2}$	$\frac{w\, p_2}{p_1(p_1 + p_2)}$			

Four minutes.

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$\min\{x_1, x_2\}$	$\frac{w}{p_1 + p_2}$	-0.48	$+0.48$	0
$x_1 x_2$	$\frac{w}{2p_1}$	0	$+0.5$	0.5
$\sqrt{x_1} + \sqrt{x_2}$	$\frac{w\, p_2}{p_1(p_1 + p_2)}$	$+0.48$	$+0.48$	0.96

Four minutes.

Good 1 normal: net $\geq$ gross. The gap of class 6 is $x_2\, \partial x_1 / \partial w$.

(optional) The Slutsky equation in elasticities

$\varepsilon_{11} = (\partial x_1/\partial p_1)\, p_1/x_1$, $\varepsilon_{1w} = (\partial x_1/\partial w)\, w/x_1$, $\varepsilon^h_{11} = (\partial h_1/\partial p_1)\, p_1/h_1$;
budget share $b_1 = p_1 x_1/w$.

(1) Show: $\varepsilon_{11} = \varepsilon^h_{11} - b_1\, \varepsilon_{1w}$.

(2) $\varepsilon^h_{11} = -0.2$. For which ε_{1w} is good 1 Giffen if $b_1 = 0.4$? If $b_1 = 0.02$?

MWG (3.G.3), p. 71; elasticities as in MWG 2.E, p. 27; budget share b_ℓ, p. 28. Invented numbers in (2).

(optional) The Slutsky equation in elasticities

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budget share $b_1 = p_1 x_1/w$.

(1) Show: $\varepsilon_{11} = \varepsilon^h_{11} - b_1\, \varepsilon_{1w}$.

Slutsky (slide 9), $\ell = k = 1$, times p_1/x_1:

$$\frac{\partial x_1}{\partial w}\, x_1\, \frac{p_1}{x_1} = \varepsilon_{1w}\, b_1$$

(2) $\varepsilon^h_{11} = -0.2$. For which ε_{1w} is good 1 Giffen if $b_1 = 0.4$? If $b_1 = 0.02$?

$\varepsilon_{1w} < -0.5$; $\varepsilon_{1w} < -10$:
Small budget share: Giffen needs an absurd ε_{1w}.

MWG (3.G.3), p. 71; elasticities as in MWG 2.E, p. 27; budget share b_ℓ, p. 28. Invented numbers in (2).

Break

Back at 11:00

The Slutsky matrix: class 6's $D_p h$, built from observable demand

$$S(p, w): \quad s_{\ell k} = \frac{\partial x_\ell}{\partial p_k} + \frac{\partial x_\ell}{\partial w} x_k$$

only Walrasian demand: observable in principle

$$S = \begin{pmatrix} -\frac{3}{2} + 3 \cdot \frac{1}{4} & 0 + 2 \cdot \frac{1}{4} \\ 0 + 3 \cdot \frac{1}{6} & -\frac{2}{3} + 2 \cdot \frac{1}{6} \end{pmatrix} = \begin{pmatrix} -\frac{3}{4} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{3} \end{pmatrix}$$

$= D_p h(p, 6)$: class 6, slide 20

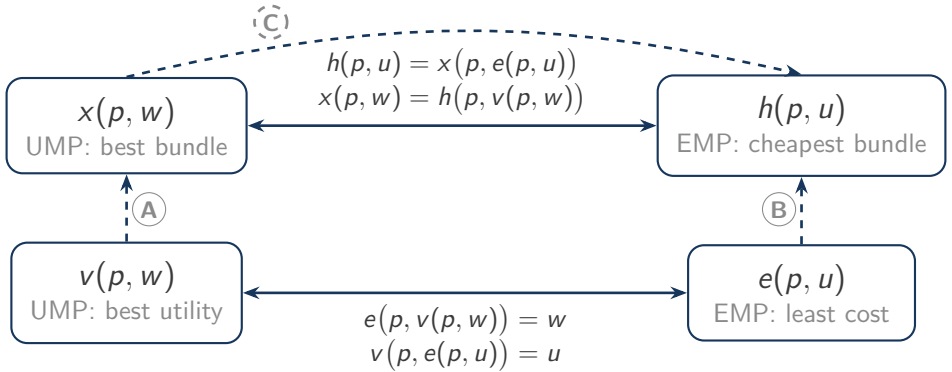
negative semidefinite symmetric $S\,p = 0$

Demand whose S is not symmetric or not negative semidefinite comes from no rational preferences.

MWG Prop. 3.G.2, pp. 69–70; 3.G, pp. 72–73: $S(p, w) = D_p h(p, u)$ (Prop. 3.G.3).

In pairs: the map of x, v, h, e — name the three derivative arrows

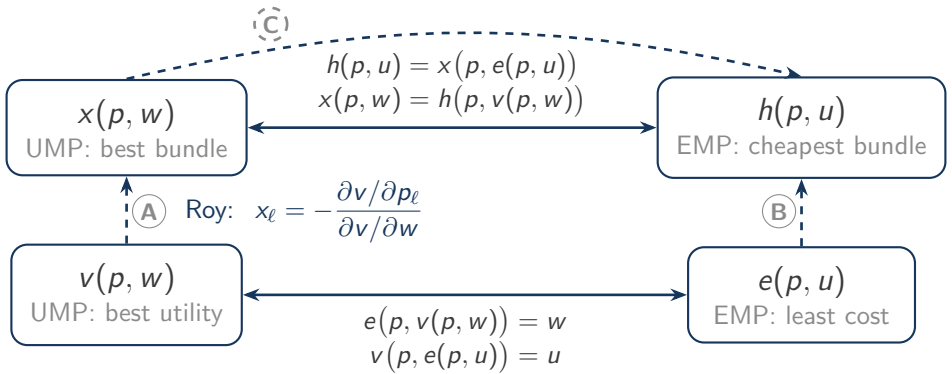
In pairs: name A, B, C — formula and name.



MWG Figure 3.G.3, p. 75 (a similar figure: Deaton and Muellbauer 1980); Props. 3.G.1, 3.G.3, 3.G.4.

In pairs: the map of x, v, h, e — name the three derivative arrows

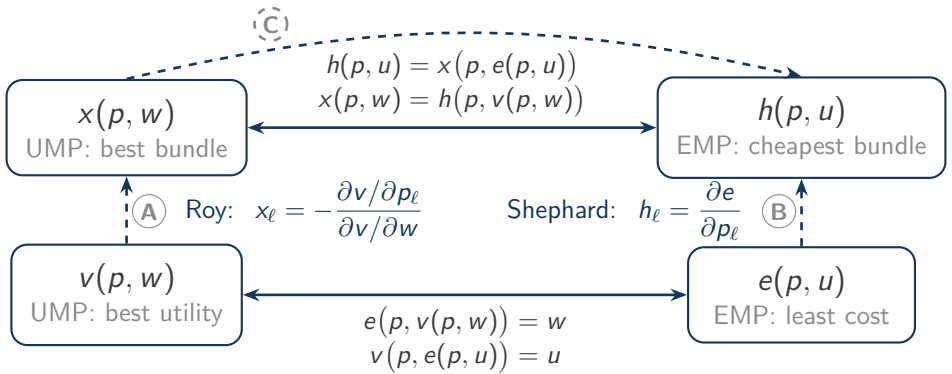
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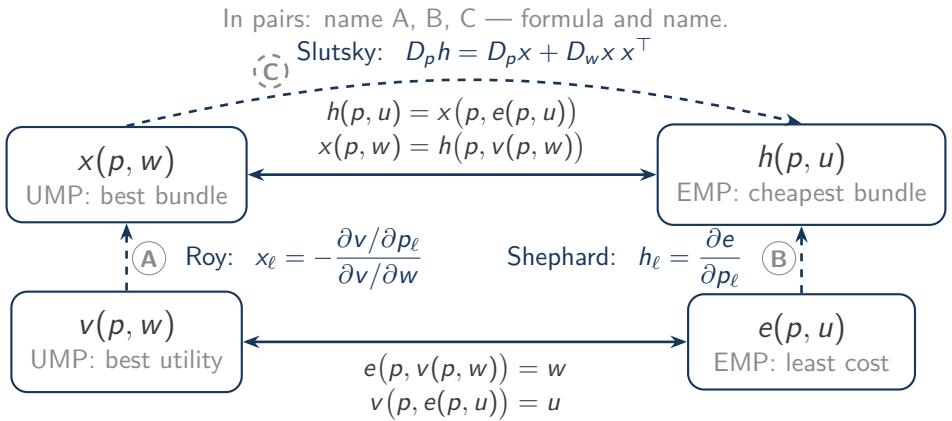
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In pairs: the map of x, v, h, e — name the three derivative arrows



Every arrow is from classes 4–7; the dashed ones are today's and last week's.

MWG Figure 3.G.3, p. 75 (a similar figure: Deaton and Muellbauer 1980); Props. 3.G.1, 3.G.3, 3.G.4.

Money measures of welfare

MWG 3.I

Price the change in euros: EV at the old prices, CV at the new

money metric: $e\big(\bar{p}, v(p, w)\big)$

utility in euros at fixed prices $\bar{p}$

equivalent variation: $\mathrm{EV} = e(p^0, u^1) - w$

at the old prices, the change in wealth as good as the price change: $v(p^0, w + \mathrm{EV}) = u^1$

compensating variation: $\mathrm{CV} = w - e(p^1, u^0)$

at the new prices, what can be taken after the change to restore u^0: $v(p^1, w - \mathrm{CV}) = u^0$

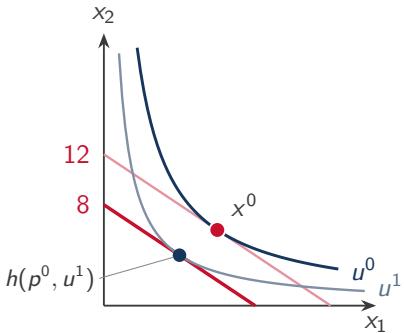
$u^0 = v(p^0, w), \quad u^1 = v(p^1, w)$

EV and CV: both negative when she loses

Both rank p^0 and p^1 correctly; their euros differ.

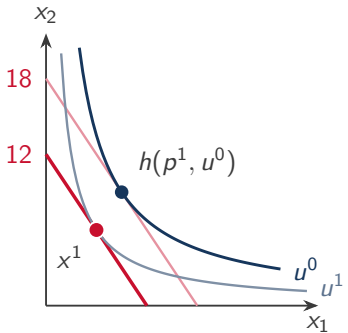
MWG 3.I, pp. 81–82: Figure 3.I.1; (3.I.1), (3.I.2), after Hicks (1939).

Good 1 to 4.5: EV $= -4$ euros, CV $= -6$ euros



EV: old prices

$e(p^0, u^1) - w = 8 - 12 = -4$



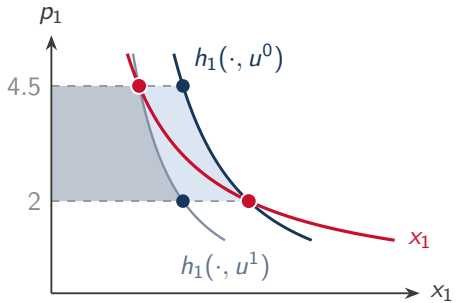
CV: new prices

$w - e(p^1, u^0) = 12 - 18 = -6$

Old prices value her loss at 4 euros, new prices at 6: EV $>$ CV, good 1 is normal.

MWG Figure 3.I.2, p. 82; $u = x_1 x_2$, $w = 12$, $p^0 = (2, 3)$, $p^1 = (4.5, 3)$.

$-\mathbf{CV} = 6$ and $-\mathbf{EV} = 4$ are areas left of Hicksian demand



$-$CV: both shades $-$EV: dark

$$-\text{CV} = \int_2^{4.5} h_1(p_1, 3, u^0)\, dp_1 = 6$$

$$-\text{EV} = \int_2^{4.5} h_1(p_1, 3, u^1)\, dp_1 = 4$$

$$\int_2^{4.5} x_1(p_1, 3, 12)\, dp_1 = 6 \ln 2.25 \approx 4.87$$

market demand: in between

Shephard turns a change in e into an area under compensated demand.

MWG 3.I, pp. 83 and 89: (3.I.3), (3.I.4), (3.I.8); Figure 3.I.3.

Buyers pay a CV, choosers take an EV; 7.12 is no wealth effect

group	in our theory	median
Buyers	CV of getting a mug: pay it and stay as well off	\$2.87
Choosers	EV of getting a mug: cash as good as the mug	\$3.12
Sellers	the choosers' choice: mug or cash	\$7.12

EV − CV: a wealth effect; for a 3-dollar mug, cents

Buyers vs choosers: as theory says. Sellers vs choosers: no wealth effect can do that; owning the mug changed its value.

Kahneman, Knetsch & Thaler (1990), JPE 98(6), p. 1339: "these results eliminate any form of income effect as an explanation".

In pairs: good 1 falls to 0.5: what is the gain worth?

$$u = x_1 x_2, \quad w = 12, \quad p^0 = (2, 3) \to p^1 = (0.5, 3).$$

question	answer
new bundle x^1 and utility u^1	
$h(p^1, u^0)$: split x_1 into substitution and wealth	
$\text{CV} = w - e(p^1, u^0)$	
$\text{EV} = e(p^0, u^1) - w$	

Five minutes.

In pairs: good 1 falls to 0.5: what is the gain worth?

$$u = x_1 x_2, \quad w = 12, \quad p^0 = (2, 3) \to p^1 = (0.5, 3).$$

question	answer
new bundle x^1 and utility u^1	$(12,\ 2)$, $u^1 = 24$
$h(p^1, u^0)$: split x_1 into substitution and wealth	
$\text{CV} = w - e(p^1, u^0)$	
$\text{EV} = e(p^0, u^1) - w$	

Five minutes.

In pairs: good 1 falls to 0.5: what is the gain worth?

$$u = x_1 x_2, \quad w = 12, \quad p^0 = (2, 3) \to p^1 = (0.5, 3).$$

question	answer
new bundle x^1 and utility u^1	$(12,\ 2)$, $u^1 = 24$
$h(p^1, u^0)$: split x_1 into substitution and wealth	$(6,\ 1)$; x_1: $3 \to 6 \to 12$ substitution $+3$, wealth $+6$
$\text{CV} = w - e(p^1, u^0)$	
$\text{EV} = e(p^0, u^1) - w$	

Five minutes.

In pairs: good 1 falls to 0.5: what is the gain worth?

$$u = x_1 x_2, \quad w = 12, \quad p^0 = (2, 3) \to p^1 = (0.5, 3).$$

question	answer
new bundle x^1 and utility u^1	$(12,\ 2)$, $u^1 = 24$
$h(p^1, u^0)$: split x_1 into substitution and wealth	$(6,\ 1)$; x_1: $3 \to 6 \to 12$ substitution $+3$, wealth $+6$
$\text{CV} = w - e(p^1, u^0)$	$12 - 6 = 6$
$\text{EV} = e(p^0, u^1) - w$	

Five minutes.

In pairs: good 1 falls to 0.5: what is the gain worth?

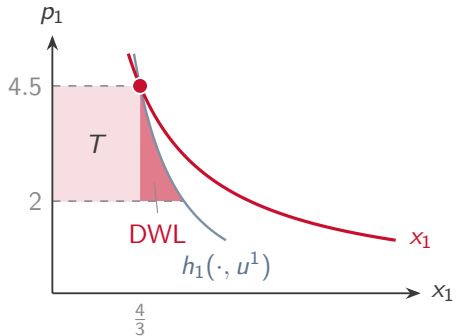
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new bundle x^1 and utility u^1	$(12,\ 2)$, $u^1 = 24$
$h(p^1, u^0)$: split x_1 into substitution and wealth	$(6,\ 1)$; x_1: $3 \to 6 \to 12$ substitution $+3$, wealth $+6$
$\text{CV} = w - e(p^1, u^0)$	$12 - 6 = 6$
$\text{EV} = e(p^0, u^1) - w$	$24 - 12 = 12$

Five minutes.

A gain: both positive, and EV $>$ CV again: good 1 is normal.

Tax good 1 by 2.5: it raises 3.33, costs her 4: deadweight loss 0.67



tax $t = 2.5$ per unit: p_1: $2 \to 4.5$

revenue $T = t\, x_1(p^1, w) = 2.5 \cdot \frac{4}{3} \approx 3.33$

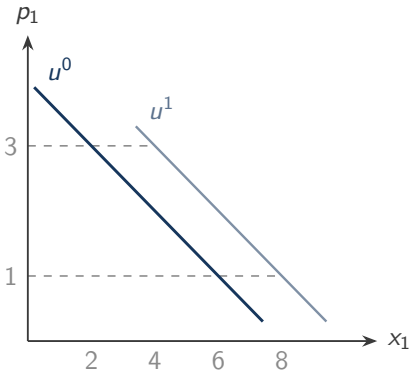
her loss $-\text{EV} = 4$

deadweight loss $= 4 - 3.33 \approx 0.67$

The tax takes more from her than it brings in: a triangle left of Hicksian demand.

MWG Example 3.I.1, pp. 84–85, (3.I.5), Figure 3.I.4(a); own numbers.

(optional) CV and EV from two compensated demand curves



p_1: $1 \to 3$; p_2, w fixed.

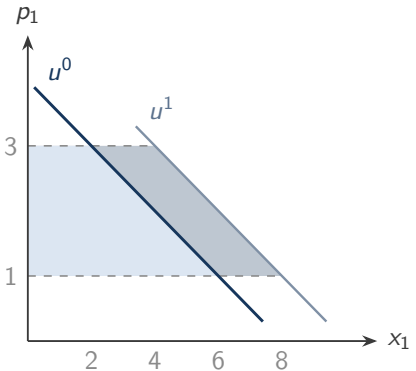
$h_1(p_1, u^0) = 8 - 2p_1$, $h_1(p_1, u^1) = 10 - 2p_1$

(1) CV and EV?

(2) Normal or inferior?
Is EV $>$ CV, as on slide 20?

MWG 3.I, p. 83: (3.I.3), (3.I.4), Figure 3.I.3. Own numbers.

(optional) CV and EV from two compensated demand curves



p_1: $1 \to 3$; p_2, w fixed.

$h_1(p_1, u^0) = 8 - 2p_1$, $h_1(p_1, u^1) = 10 - 2p_1$

(1) CV and EV?

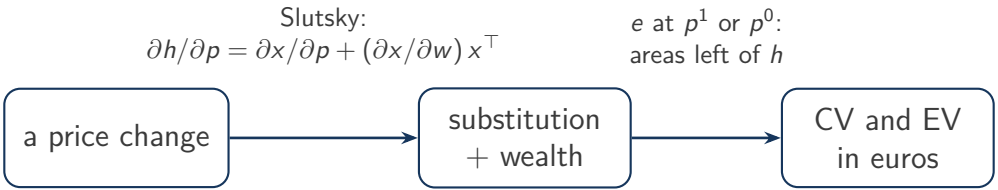
CV $= -8$ (light), EV $= -12$ (both shades)

(2) Normal or inferior?
Is EV $>$ CV, as on slide 20?

Inferior: more of it at the lower u^1.
No: EV $<$ CV.

MWG 3.I, p. 83: (3.I.3), (3.I.4), Figure 3.I.3. Own numbers.

Today in one line



Wednesday: Uncertainty I: lotteries, the independence axiom, the expected utility theorem and the uniqueness of the vNM utility function.

Money lotteries, risk aversion, certainty equivalent, risk premium.

Problem set 2: due at the start of class, Wednesday 21 October — on paper, one copy per group.

Read MWG 6.B–6.C (or Muñoz-Garcia ch. 5) for Wednesday 21 October.